

SAVITHRI MARATI

QA Engineer | DevOps & Cloud Infrastructure (AWS)

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PROFESSIONAL SUMMARY

QA Engineer with 3 years 7 months of experience spanning quality assurance and DevOps/cloud infrastructure engineering on AWS. Experienced in building CI/CD pipelines (Jenkins, GitHub Actions), provisioning infrastructure as code with Terraform, and operating Kubernetes (EKS) workloads in production. Skilled in functional and regression testing, GitOps workflows with Argo CD, monitoring and observability with Prometheus and Grafana, and using AI-assisted tooling (Claude with MCP) to accelerate test case authoring. Reduced EC2/S3 cloud spend by 20% through rightsizing and lifecycle policy optimization, and cut MLOps deployment cycle time by 60%. Experienced in mentoring junior engineers on CI/CD and cloud best practices.

CORE COMPETENCIES

Cloud & Infrastructure: AWS (EC2, S3, IAM, VPC, RDS, Route 53, CloudWatch), Terraform, Docker, Kubernetes (EKS), Helm, Ansible

CI/CD & DevOps: Jenkins, GitHub Actions, Argo CD, GitOps, Git, Infrastructure as Code (IaC)

QA & Testing: Functional Testing, Regression Testing, Smoke Testing, Test Case Design, Defect Triage, AI-Assisted Test Authoring (Claude, MCP)

MLOps: MLflow, DVC, KServe, Model Deployment Pipelines, Continuous Training

Monitoring & Logging: Prometheus, Grafana, EFK Stack, Dynatrace (exposure), CloudWatch

Databases: MySQL, PostgreSQL, MongoDB, Redis

Scripting & Languages: Python, Bash

Security & Code Quality: SonarQube, Trivy, IAM Least-Privilege Policy Design

Additional Exposure: OpenShift

PROFESSIONAL EXPERIENCE

Associate QA Engineer

Apr 2024 – Present (2 yrs 6 mos)

ScleraVDMS Private Limited | Mangaluru, India

- Built and maintained **CI/CD pipelines** using **Jenkins** and **GitHub Actions**, reducing manual deployment steps and increasing release frequency across multiple services.
- Provisioned **AWS infrastructure** (EC2, S3, RDS, IAM, VPC, EKS, Route 53, CloudWatch) using **Terraform**, enabling repeatable, auditable deployments across dev, staging, and production environments.
- Containerized microservices with **Docker** and deployed to **AWS EKS**, supporting zero-downtime rolling updates; contributed to an ongoing monolith-to-microservices migration.
- Implemented **GitOps** workflows with **Argo CD**, enabling version-controlled deployments and faster rollback during production incidents.
- Reduced cloud infrastructure costs by approximately **20%** through EC2 rightsizing, S3 lifecycle policies, and Reserved Instance planning.
- Performed functional, regression, and smoke testing for microservices prior to release, coordinating with development teams to triage and log defects used **Claude with MCP (Model Context Protocol)** to accelerate test case authoring for regression coverage.
- Built and operated an end-to-end **MLOps pipeline** (MLflow, DVC, Docker, KServe), reducing model deployment cycle time by approximately **60%**.
- Built monitoring dashboards with **Prometheus** and **Grafana** and maintained centralized logging with the **EFK stack**, improving P1/P2 incident response and cross-service troubleshooting.
- Automated routine infrastructure health checks with **Bash** and **Python**, and configured **IAM** roles and security groups following least-privilege principles.
- Managed **Helm** chart deployments for standardized packaging, maintained **Ansible** playbooks for server provisioning, and mentored new team members on CI/CD and cloud practices.

Junior System Engineer

Mar 2023 – Mar 2024 (1 yr 1 mo)

Access Research Labs (ARL) | Mangaluru, India

- Managed and monitored **Linux**-based server infrastructure, ensuring system stability, uptime, and performance across production and staging environments.
- Assisted in provisioning **AWS** cloud resources (EC2, S3, IAM) and supported initial **Terraform**-based infrastructure automation.
- Set up and maintained **CI/CD pipelines** using **Jenkins**, supporting faster and more reliable application deployments.
- Performed routine system health checks, log analysis, and incident triage, resolving **P1/P2** issues to maintain **SLA** compliance.
- Collaborated with development teams on **GitFlow** branching strategies, code review gates, and quality checks within CI/CD workflows.
- Containerized applications using **Docker** and assisted with initial **Kubernetes** deployment configurations for microservices.
- Authored and maintained internal runbooks and documentation for system administration and incident response.

KEY PROJECTS

Production-Grade MLOps Pipeline (Customer Churn Prediction Model)

- Designed an end-to-end ML pipeline with CI/CD integration, dataset versioning via **DVC**, and **Kubernetes**-based model serving.
- Enabled continuous training and deployment using **Argo CD** and **KServe**, reducing deployment cycle time by approximately **60%**.

Technologies: Python, Docker, Kubernetes, Argo CD, KServe, MLflow, DVC, GitHub Actions

Asset Data Collection Platform (Three-Tier Architecture)

- Designed and deployed a three-tier microservices architecture (React frontend, Node.js backend, PostgreSQL database) using **Docker** and **Kubernetes**.
- Built **Jenkins** CI/CD pipelines and **Terraform** IaC modules for automated infrastructure provisioning and application delivery.
- Implemented a GitOps workflow with **Argo CD** and **Helm**, integrated **Prometheus/Grafana** monitoring, and centralized logging with the **EFK stack**.

Technologies: AWS EKS, Docker, Kubernetes, Jenkins, Terraform, Argo CD, Helm, Prometheus, Grafana, EFK Stack

AWS Cloud Cost Optimization

- Rightsized **EC2** instances based on utilization metrics, eliminating overprovisioned compute and reducing monthly EC2 spend.
- Wrote an **AWS Lambda** function to automatically identify and delete stale/orphaned **EBS snapshots**, reducing unnecessary storage costs.
- Implemented **S3 lifecycle policies** to transition and expire objects based on access patterns, cutting storage costs across environments.

Technologies: AWS EC2, Lambda, EBS, S3, Python, CloudWatch

Inventory Management System (Microservices)

- Handled deployment of a multi-service microservices platform comprising 4 services: **Authentication and User services (Java)**, a **Notification service (Go)**, and a **Data Processing service (Python)**.
- Containerized and deployed each service independently, coordinating releases and monitoring across a polyglot microservices stack.

Technologies: Java, Go, Python, Docker, Kubernetes

EDUCATION

Bachelor of Engineering — Electronics and Communication Engineering

Canara Engineering College | 2019 – 2023 | CGPA: 8.7 / 10.0